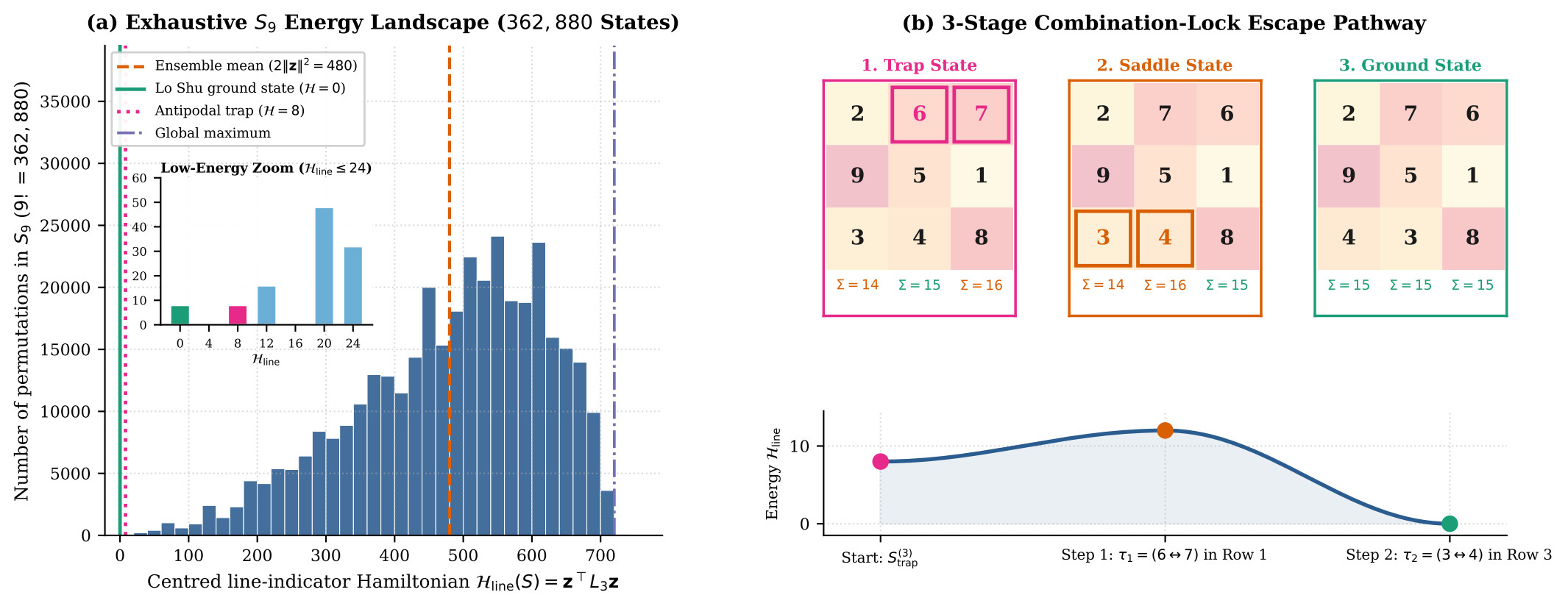

(a) Exhaustive S_9 Energy Landscape (362,880 States)
Ensemble mean ($2\|\mathbf{z}\|^2 = 480$)
Lo Shu ground state ($\mathcal{H} = 0$)
Antipodal trap ($\mathcal{H} = 8$)
Global maximum
Low-Energy Zoom ($\mathcal{H}_{\text{line}} \leq 24$)
$\mathcal{H}_{\text{line}}$
Number of permutations in S_9 ($9! = 362,880$)
Centred line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S) = \mathbf{z}^\top L_3 \mathbf{z}$
(b) 3-Stage Combination-Lock Escape Pathway
1. Trap State
2. Saddle State
3. Ground State
Σ = 14 Σ = 15 Σ = 16
Σ = 14 Σ = 16 Σ = 15
Σ = 15 Σ = 15 Σ = 15
Energy $\mathcal{H}_{\text{line}}$
Start: $S^{(3)}_{\text{trap}}$
Step 1: $\tau_1 = (6 \leftrightarrow 7)$ in Row 1
Step 2: $\tau_2 = (3 \leftrightarrow 4)$ in Row 3